**Phase 1: Basic logic aur digit**

1. Sample Question
2. FizzBuzz
3. Is Rectangle?
4. Palindrome Integer
5. Reverse integer

**Phase 2: Base conversion**

6. Binary Representation
7. Excel Column Number
8. Excel Column Title

**Phase 3: Divisibility aur factors**

9. Greatest Common Divisor
10. Divisible by 60
11. All Factors
12. Verify Prime
13. Trailing Zeros in Factorial

**Phase 4: Sequences aur Sieve**

14. Find Nth Fibonacci
15. Prime Numbers
16. Powerful Divisors

**Phase 5: Observation aur pattern**

17. Total Moves For Bishop!
18. Distribute in Circle!
19. Step by Step
20. Power Of Two Integers

**Phase 6: Sieve + GCD combine**

21. Prime Sum
22. Largest Coprime Divisor

**Phase 7: Bits aur Combinatorics**

23. Sum of pairwise Hamming Distance
24. Grid Unique Paths
25. K-th Permutation

26. Sorted Permutation Rank
27. City Tour

**Phase 8: Sabse mushkil**
28. Sorted Permutation Rank with Repeats
29. Next Smallest Palindrome!

**Phase 0: Primers (sabse pehle)**

1. ARRAY_IMPL1
2. ARRAY_2D
3. ARRAY_BUG

**Phase 1: Simple traversal aur simulation**
4. Array Sum
5. Move Zeroes
6. Segregate 0s and 1s in an array
7. Max Min
8. Add One To Number
9. Make equal elements Array

**Phase 2: Pehle se sochne wala logic (prefix, suffix, running value)**
10. Max Sum Contiguous Subarray
11. Max Non Negative SubArray
12. Pick from both sides!
13. Perfect Peak of Array
14. Balance Array
15. Maximum Absolute Difference
16. Min Steps in Infinite Grid

**Phase 3: Sorting aur arrangement**
17. Sort array with squares!
18. Merge Overlapping Intervals
19. Merge Intervals
20. Largest Number
21. Maximum Sum Triplet
22. Partitions

**Phase 4: 2D matrix aur patterns**

23. Pascal Triangle
24. Kth Row of Pascal's Triangle
25. Anti Diagonals
26. Spiral Order Matrix I
27. Spiral Order Matrix II
28. Rotate Matrix

**Phase 5: Permutation aur arrangement logic**

29. Next Permutation
30. Find Permutation

**Phase 6: Ab tak ke concepts ka advanced use**

31. Large Factorial
32. Minimum Lights to Activate
33. Triplets with Sum between given range
34. Maximum Area of Triangle!

**Phase 7: Sabse mushkil**

35. Find Duplicate in Array
36. Maximum Consecutive Gap
37. Flip

**Phase 1: Basic binary search (lower bound / upper bound)**

1. Sorted Insert Position
2. Smaller or equal elements
3. Count Element Occurence
4. Search for a Range
5. Matrix Search

**Phase 2: Modified binary search (sorted ka twist)**

6. Rotated Array
7. Search in Bitonic Array!

**Phase 3: Search on answer (answer pe binary search)**

8. Square Root of Integer
9. WoodCutting Made Easy!

10. Capacity To Ship Packages Within B Days
11. Allocate Books
12. Painter's Partition Problem

**Phase 4: Sabse mushkil**

13. Matrix Median
14. Red Zone

**Phase 1: Basic string traversal**

1. Palindrome String
2. Remove Consecutive Characters
3. String And Its Frequency
4. Vowel and Consonant Substrings!
5. Self Permutation
6. Serialize
7. Deserialize

**Phase 2: Do strings ko compare karna**

8. Longest Common Prefix
9. Bulls and Cows
10. Count And Say

**Phase 3: Substring aur search**

11. Amazing Subarrays
12. Implement StrStr
13. Stringoholics

**Phase 4: String math (number systems ka use)**

14. Roman To Integer
15. Integer To Roman
16. Add Binary Strings
17. Salutes
18. Multiply Strings
19. Power of 2

**Phase 5: Palindrome tricks**

20. Convert to Palindrome
21. Minimum Parantheses!
22. Minimum Characters required to make a String Palindromic
23. Minimum Appends for Palindrome!
24. Longest Palindromic Substring

**Phase 1: Bits ki basic pehchaan**

1. Number of 1 Bits
2. Trailing Zeroes
3. Reverse Bits

**Phase 2: Bits ko count karna**

4. Different Bits Sum Pairwise
5. Count Total Set Bits

**Phase 3: XOR ka use**

6. Min XOR value
7. XOR-ing the Subarrays!

**Phase 4: Bits + logic ka combine**

8. Palindromic Binary Representation
9. Divide Integer.

**Phase 1: Inplace updates (two pointers ki basic aadat)**

1. Remove Element from Array
2. Remove Duplicates from Sorted Array
3. Remove Duplicates from Sorted Array II
4. Sort by Color

**Phase 2: Do arrays ko saath chalana**

5. Merge Two Sorted Lists II
6. Intersection Of Sorted Arrays

**Phase 3: Pairs dhundhna (sorting + two pointers)**

7. Pair With Given Difference
8. Diffk
9. 3 Sum
10. Container With Most Water
11. Counting Triangles

**Phase 4: Window aur counting wale tricks**

12. Max Continuous Series of 1s
13. Maximum Ones After Modification
14. Counting Subarrays!
15. Subarrays with distinct integers!

**Phase 5: Sabse mushkil**

16. Array 3 Pointers

**Phase 1: Recursion ki basic**

1. Modular Expression
2. Reverse Link List Recursion

**Phase 2: Subsets (include / exclude ka idea)**

3. All Possible Combinations
4. Subset
5. Subsets II
6. Combinations
7. Combination Sum
8. Combination Sum II

**Phase 3: Permutations**

9. Permutations
10. Maximal String
11. Kth Permutation Sequence

**Phase 4: Build karte jao (choices + conditions)**

12. Letter Phone
13. Generate all Parentheses II
14. Palindrome Partitioning
15. Gray Code

**Phase 5: Game solving (sabse mushkil)**

16. NQueens
17. Sudoku

**Phase 1: Hash ki basic (frequency aur lookup)**

1. First Repeating element
2. Colorful Number
3. Check Palindrome!
4. 2 Sum
5. Pairs With Given Xor
6. Diffk II

**Phase 2: Prefix sum + hash**

7. Largest Continuous Sequence Zero Sum
8. Longest Subarray Length
9. Subarray with given XOR
10. Subarray with B odd numbers
11. An Increment Problem

**Phase 3: Key banana (hash ki key khud design karna)**

12. Anagrams
13. Two out of Three
14. Valid Sudoku
15. Equal
16. 4 Sum

**Phase 4: Hash + two pointers (sliding window)**

17. Longest Substring Without Repeat
18. Window String

**Phase 5: Maths aur hashing**

19. Fraction
20. Points on the Straight Line

**Phase 6: Sabse mushkil**

21. Copy List
22. Substring Concatenation

**Level 1: Loops, conditions and traversal**

*Concept: reading input, looping, and handling edge cases*.

- Sample Question (Math)
- FizzBuzz (Math)
- Is Rectangle? (Math)
- Palindrome Integer (Math)
- Reverse integer (Math)
- ARRAY_IMPL1 (Array)
- ARRAY_2D (Array)
- ARRAY_BUG (Array)
- Array Sum (Array)
- Max Min (Array)
- Move Zeroes (Array)
- Segregate 0s and 1s in an array (Array)
- Add One To Number (Array)
- Make equal elements Array (Array)
- Palindrome String (String)
- Remove Consecutive Characters (String)
- String And Its Frequency (String)
- Vowel and Consonant Substrings! (String)
- Self Permutation (String)
- Serialize (String)
- Deserialize (String)
- Number of 1 Bits (Bit)
- Trailing Zeroes (Bit)
- Reverse Bits (Bit)

**Level 2: Number systems, factors and in-place updates**

*Concept: working with digits and bases, finding divisors, and changing an array without extra space.*

- Binary Representation (Math)
- Excel Column Number (Math)
- Excel Column Title (Math)
- Greatest Common Divisor (Math)
- Divisible by 60 (Math)
- All Factors (Math)
- Verify Prime (Math)
- Trailing Zeros in Factorial (Math)
- Remove Element from Array (Two Pointers)
- Remove Duplicates from Sorted Array (Two Pointers)
- Remove Duplicates from Sorted Array II (Two Pointers)
- Sort by Color (Two Pointers)
- Longest Common Prefix (String)
- Bulls and Cows (String)
- Count And Say (String)
- Modular Expression (Backtracking)
- Reverse Link List Recursion (Backtracking)

**Level 3: Searching smartly (binary search, hash lookup, sorted arrays)**

*Concept: stop checking everything. Use sorted order or a lookup table to cut the work.*

- Sorted Insert Position (Binary Search)
- Smaller or equal elements (Binary Search)
- Count Element Occurence (Binary Search)
- Search for a Range (Binary Search)

- Matrix Search (Binary Search)
- First Repeating element (Hashing)
- Colorful Number (Hashing)
- Check Palindrome! (Hashing)
- 2 Sum (Hashing)
- Pairs With Given Xor (Hashing)
- Diffk II (Hashing)
- Merge Two Sorted Lists II (Two Pointers)
- Intersection Of Sorted Arrays (Two Pointers)
- Different Bits Sum Pairwise (Bit)
- Count Total Set Bits (Bit)

**Level 4: Prefix, suffix and running values**

*Concept: remember what you have already computed instead of recomputing it*.

- Max Sum Contiguous Subarray (Array)
- Max Non Negative SubArray (Array)
- Pick from both sides! (Array)
- Perfect Peak of Array (Array)
- Balance Array (Array)
- Maximum Absolute Difference (Array)
- Min Steps in Infinite Grid (Array)
- Largest Continuous Sequence Zero Sum (Hashing)
- Longest Subarray Length (Hashing)
- Subarray with given XOR (Hashing)
- Subarray with B odd numbers (Hashing)
- An Increment Problem (Hashing)
- Min XOR value (Bit)

- XOR-ing the Subarrays! (Bit)

**Level 5: Sorting, pairs and string searching**

*Concept: sort first, then use two pointers or merging to find pairs and groups. Also search for a pattern inside a string.*

- Pair With Given Difference (Two Pointers)
- Diffk (Two Pointers)
- 3 Sum (Two Pointers)
- Container With Most Water (Two Pointers)
- Counting Triangles (Two Pointers)
- Sort array with squares! (Array)
- Merge Overlapping Intervals (Array)
- Merge Intervals (Array)
- Largest Number (Array)
- Maximum Sum Triplet (Array)
- Partitions (Array)
- Rotated Array (Binary Search)
- Search in Bitonic Array! (Binary Search)
- Amazing Subarrays (String)
- Implement StrStr (String)
- Stringoholics (String)

**Level 6: Search on the answer, windows and number sequences**

*Concept: binary search on the answer itself, and sliding windows that grow and shrink.*

- Square Root of Integer (Binary Search)
- WoodCutting Made Easy! (Binary Search)
- Capacity To Ship Packages Within B Days (Binary Search)
- Allocate Books (Binary Search)

- Painter's Partition Problem (Binary Search)
- Max Continuous Series of 1s (Two Pointers)
- Maximum Ones After Modification (Two Pointers)
- Counting Subarrays! (Two Pointers)
- Subarrays with distinct integers! (Two Pointers)
- Longest Substring Without Repeat (Hashing)
- Window String (Hashing)
- Find Nth Fibonacci (Math)
- Prime Numbers (Math)
- Powerful Divisors (Math)

**Level 7: Recursion and backtracking (choices and decision trees)**

*Concept: at every step choose or skip, explore, and undo the choice.*

- All Possible Combinations (Backtracking)
- Subset (Backtracking)
- Subsets II (Backtracking)
- Combinations (Backtracking)
- Combination Sum (Backtracking)
- Combination Sum II (Backtracking)
- Permutations (Backtracking)
- Maximal String (Backtracking)
- Kth Permutation Sequence (Backtracking)
- Letter Phone (Backtracking)
- Generate all Parentheses II (Backtracking)
- Palindrome Partitioning (Backtracking)
- Gray Code (Backtracking)

**Level 8: Matrix handling and pattern observation**

*Concept: find the pattern or formula behind the problem instead of simulating it.*

- Pascal Triangle (Array)
- Kth Row of Pascal's Triangle (Array)
- Anti Diagonals (Array)
- Spiral Order Matrix I (Array)
- Spiral Order Matrix II (Array)
- Rotate Matrix (Array)
- Total Moves For Bishop! (Math)
- Distribute in Circle! (Math)
- Step by Step (Math)
- Power Of Two Integers (Math)

**Level 9: String math, bits and palindromes**

*Concept: do arithmetic on strings and bits, and build palindromes.*

- Roman To Integer (String)
- Integer To Roman (String)
- Add Binary Strings (String)
- Salutes (String)
- Multiply Strings (String)
- Power of 2 (String)
- Palindromic Binary Representation (Bit)
- Divide Integers (Bit)
- Convert to Palindrome (String)
- Minimum Parantheses! (String)
- Minimum Characters required to make a String Palindromic (String)
- Minimum Appends for Palindrome! (String)
- Longest Palindromic Substring (String)

**Level 10: Designing your own hash key and combining ideas**

*Concept: decide what to store as the key, and combine earlier tools such as sieve, GCD, greedy and permutations.*

- Anagrams (Hashing)
- Two out of Three (Hashing)
- Valid Sudoku (Hashing)
- Equal (Hashing)
- 4 Sum (Hashing)
- Fraction (Hashing)
- Points on the Straight Line (Hashing)
- Prime Sum (Math)
- Largest Coprime Divisor (Math)
- Next Permutation (Array)
- Find Permutation (Array)
- Large Factorial (Array)
- Minimum Lights to Activate (Array)
- Triplets with Sum between given range (Array)
- Maximum Area of Triangle! (Array)

**Level 11: Combinatorics and counting**

*Concept: count arrangements and paths using formulas and bit tricks.*

- Sum of pairwise Hamming Distance (Math)
- Grid Unique Paths (Math)
- K-th Permutation (Math)
- Sorted Permutation Rank (Math)
- City Tour (Math)

**Level 12: Hardest questions**

*Concept: combine everything and prove your solution works.*

- Find Duplicate in Array (Array)
- Maximum Consecutive Gap (Array)
- Flip (Array)
- Matrix Median (Binary Search)
- Red Zone (Binary Search)
- Array 3 Pointers (Two Pointers)
- NQueens (Backtracking)
- Sudoku (Backtracking)
- Copy List (Hashing)
- Substring Concatenation (Hashing)
- Sorted Permutation Rank with Repeats (Math)
- Next Smallest Palindrome! (Math)